

Markscheme

Mathematics: analysis and approaches
Practice question bank

128. (a)

Consider the equation $(x + yi)(2 - i) = 8 + i$, where x and y are real.

Find the value of x and the value of y .

expand: $2x + y + (2y - x)i = 8 + i$ (M1)

equate real and imaginary parts: $2x + y = 8$ and $2y - x = 1$ A1

solve simultaneously: $y = 8 - 2x \Rightarrow 2(8 - 2x) - x = 1$ (M1)

$x = 3, y = 2$ A1

[5 marks]